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Trigram Model

Extend the bigram model to consider *two* words of context instead of one, leading to better text generation.

You will need

- same as *Basic Training* lesson
- additional paper for the three-column model
- pen, paper & dice as per *Basic Generation*

Your goal

To train a trigram language model (a table this time, not a grid like your bigram model from *Basic Training*) and use it to generate text. **Stretch goal:** train on more data, or generate more text.

Key idea

More context leads to better predictions. A trigram model considers two previous words instead of one, demonstrating the trade-off between context length and data requirements that shapes all language models.

Algorithm (training)

1. create a four-column table (see example on right)
2. **extract all word triples:** for each (overlapping) word 1/word 2/word 3 “triple” in your text increment the **count** column for that triple, or create a new row if it’s a triple you’ve never seen before and set the count to 1 (note: row order doesn’t matter)

Example (training)

After the first four words (see spot run .) the model is:

word 1	word 2	word 3	count
see	spot	run	
spot	run	.	

Note: the order of the rows doesn’t matter, so you can re-order to group them by **word 1** if that helps.

Algorithm (generation)

1. pick any row from your table; write down **word 1** and **word 2** as your starting words
2. find *all* rows where **word 1** and **word 2** are exact matches for your two starting words, and make note of their **count** columns
3. as per *Basic Generation* roll a d10 weighted by the counts and select the **word 3** associated with the chosen row
4. move along by *one* word (so **word 2** becomes your new **word 1** and **word 3** becomes your new **word 2**) and repeat from step 2

After the full text (see spot run . see spot jump .) the model is:

word 1	word 2	word 3	count
see	spot	run	
spot	run	.	
run	.	see	
.	see	spot	
see	spot	jump	
spot	jump	.	

Example (generation)

1. from the table above, choose **see** (**word 1**) and **spot** (**word 2**) as your starting words
2. find all rows with **word 1** = **see** and **word 2** = **spot**; in this case rows 1 and 5 (both have *count* == 1)
3. roll a d10 and write down the **word 3** from the row chosen by the dice roll
4. move along by *one* word (so **word 1** is **spot** and **word 2** is either **run** or **jump** depending on your dice roll) and repeat from step 2